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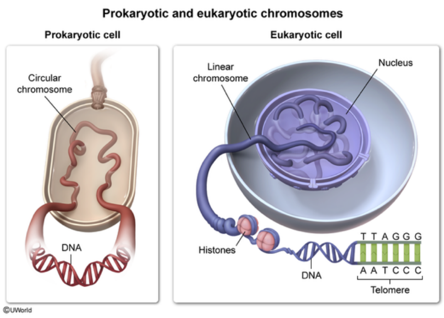
Based on the passage, which of the following best explains how mitochondrial genes differed from eukaryotic genes prior to gene transfer? Mitochondrial genes were:

- A. interspersed with noncoding sequences. [13%]
- ✔ B. located on a chromosome without telomeres. [41%]
- C. present on a single-stranded DNA genome. [39%]
- D. associated with histones. [4%]

Correct
41% answered correctly

Explanation:



The passage states mitochondria originated as aerobic prokaryotes that were engulfed by primitive anaerobic eukaryotes. The prokaryotic organisms (mitochondria) then began living inside the larger eukaryotic host, establishing a mutually beneficial symbiotic relationship.

This endosymbiotic theory of eukaryotic evolution is widely accepted because, like bacteria, mitochondria have their own genome (ie, mitochondrial DNA) and possess an independent system for transcription and translation distinct from that of the nuclear eukaryotic genome.

Prior to gene transfer, mitochondrial genes were **prokaryotic in structure** and located in the nucleoid region of the cell. According to the passage, a partial transfer of mitochondrial genes to the eukaryotic nucleus occurred. These transferred mitochondrial genes became identical in structure to nuclear genes, but the residual mitochondrial genes remained in prokaryotic form. This characteristic means that, like bacteria, mitochondrial genetic information is still contained within a **circular** chromosome of **double-stranded DNA** that has **no telomeres** and is not associated with histone proteins.

In contrast, eukaryotic genes are located on double-stranded linear chromosomes capped by telomeres, which are sequences of repetitive nucleotides located at each end of a chromosome. These regions shorten after each round of replication, protect the DNA from deterioration, and prevent the tip of chromosomes from fusing with adjacent chromosomes.

(Choice A) Introns are found in eukaryotic DNA, not prokaryotic DNA. Introns are DNA regions that do not code for proteins but are removed by the spliceosome during formation of a mature messenger RNA molecule.

(Choice C) Although prokaryotes have only a single DNA *chromosome*, the DNA of *both* prokaryotic and eukaryotic genomes is still *double-stranded*, not single-stranded.

(Choice D) Histone proteins are restricted to archaeal cells and the nuclei of eukaryotic cells but are not found in bacteria. Histones associate with DNA to facilitate its organization and packaging into chromosomes. Bacterial genes would not have been associated with histones *prior to gene transfer* to eukaryotes. Archaeal genes do not differ from eukaryotic genes on the basis of histone association.

Educational objective:
In the nucleoid region of bacterial prokaryotic cells, double-stranded DNA is condensed into a circular chromosome that has no telomeres or associated histones. In contrast, eukaryotic cells package their histone-wrapped, double-stranded DNA into linear chromosomes with ends capped by telomeres to prevent DNA from unraveling.

Time spent:0 sec	**QID**:400782
Subject:	**System:**
Biology	Cellular Biology

Eukaryotic organelles such as mitochondria and chloroplasts are believed to be relics of formerly free-living prokaryotes. The transition from a hypoxic (low O_2) to an oxic atmosphere (21% O_2) is said to have enabled primitive eukaryotic anaerobes to engulf ancient aerobic prokaryotes and consequently acquire the ability to produce energy through oxidative phosphorylation. This endosymbiotic theory of eukaryotic evolution also postulates that endosymbiosis resulted in larger eukaryotic genomes, which originated from the partial transfer of mitochondrial genes to the nuclear genome. On integration into the host genome, mitochondria-derived genes became indistinguishable from the original nuclear genes.

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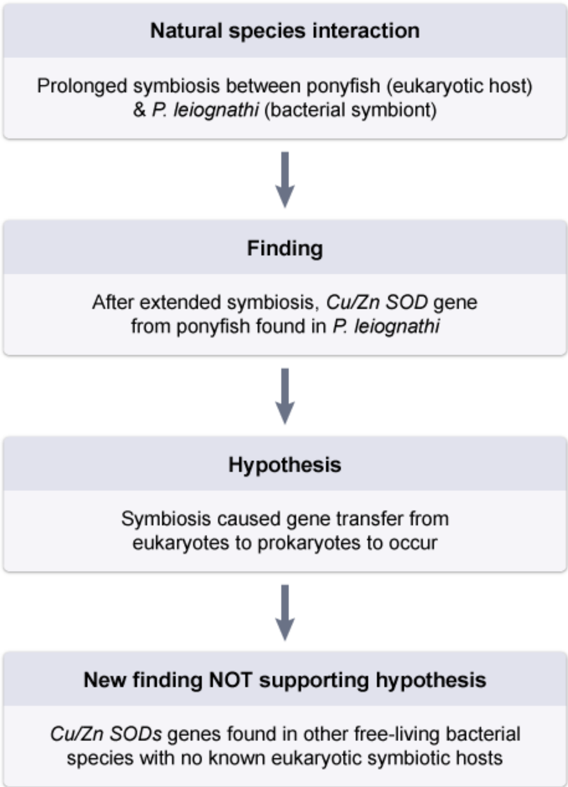
Which observation would NOT support the hypothesis of gene transfer from eukaryotes to prokaryotes described in the passage?

- A. Discovering no difference between the gene sequences of the *P. leiognathi* and ponyfish SODs [4%]
- B. Discovering *P. leiognathi* and ponyfish SODs have a similar amino acid sequence in their noncatalytic domains [2%]
- C. Discovering Cu/Zn SODs in other free-living bacterial species with no known eukaryotic symbiotic hosts [87%]
- D. Discovering Cu/Zn SODs in other free-living bacterial species with known eukaryotic symbiotic hosts [4%]

Correct
87% answered correctly

Explanation:



According to the passage, scientists proposed that a **prolonged symbiosis** period could induce **gene transfer** from eukaryotes to **prokaryotes**. In the given example, the symbiont (ie, *P. leiognathi*) was believed to have acquired the gene coding for Cu/Zn SOD from its eukaryotic host (ie, ponyfish) after an extended period of symbiosis. The passage states that Cu/Zn SOD is typically found in the cytosol of *eukaryotes*; therefore, scientists concluded that SODs found in *P. leiognathi* were most likely acquired from ponyfish. However, the discovery of **Cu/Zn SODs** in **other free-living bacterial species** with **no known eukaryotic symbiotic hosts** would **argue against** the **hypothesis** of eukaryotic to prokaryotic gene transfer due to symbiosis.

(Choices A and B) The amino acid residues in the catalytic domain (active site) of an enzyme are involved in protein function because they catalyze the enzymatic reaction. In contrast, residues in noncatalytic domains constitute a large percentage of the gene (and protein) sequence but do not contribute primarily to catalytic function. Because sequence similarity indicates common evolutionary ancestry, the hypothesis of gene transfer from eukaryotes to prokaryotes would be *supported* by the discovery of a similar SOD gene sequence between *P. leiognathi* and ponyfish or a similar amino acid sequence in the noncatalytic domains of *P. leiognathi* and ponyfish SODs.

(Choice D) The discovery of Cu/Zn SODs in other free-living bacterial species with known eukaryotic symbiotic hosts *reinforces* the hypothesis of eukaryotic to prokaryotic gene transfer.

Educational objective:
The endosymbiotic theory explains how primitive eukaryotic anaerobes engulfed ancient aerobic prokaryotes and consequently acquired the ability to produce energy through oxidative phosphorylation.

Time spent:0 sec	**QID**:400783
Subject:	**System:**
Biology	Cellular Biology

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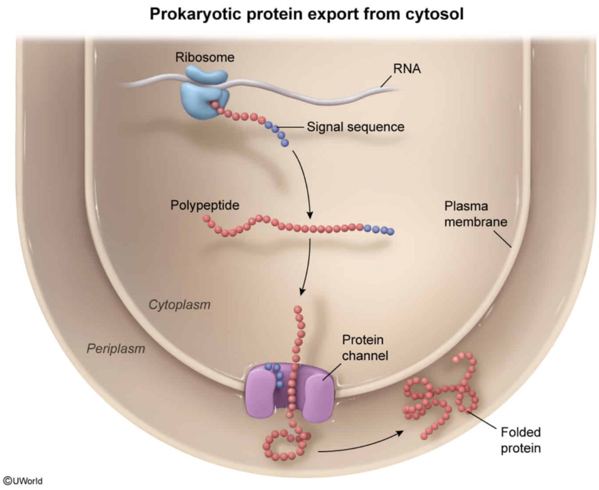
SODs are antioxidant enzymes that serve as the cell's first line of defense against reactive oxygen species (ROS). ROS produced by the electron transport chain damage proteins by oxidizing amino acid residues and metal ions on prosthetic groups but can accumulate during times of biochemical and environmental stress. Superoxide (O_2^-) radicals, a form of ROS, are sequestered by SODs and converted into less toxic hydrogen peroxide (H_2O_2) and O_2 gas.

SOD mRNA in *P. leiognathi* encodes a signal sequence that directs the transport of the SOD protein for secretion. This signal sequence will direct SOD proteins to which structure in *P. leiognathi*?

- A. Smooth endoplasmic reticulum [7%]
- ✔ B. Plasma membrane [41%]
- C. Mitochondrial outer membrane [6%]
- D. Golgi body [44%]

Correct
40% answered correctly

Explanation:



According to the question, SOD mRNA in *P. leiognathi* encodes a signal sequence that directs the transport of the SOD protein to the extracellular environment. Signal sequences are short amino acid sequences typically located at the N-terminus of a polypeptide, and are used by both eukaryotic and prokaryotic cells to direct proteins to precise destinations within the cell.

P. leiognathi is a unicellular prokaryotic organism, and therefore **lacks membrane-bound organelles** such as the Golgi body and the smooth endoplasmic reticulum (SER) present in eukaryotes. As a result, in *P. leiognathi* this **signal sequence guides** the **protein destined for secretion** (ie, SOD) directly to **protein channels** found on the **plasma membrane**. The channel contains a recognition sequence that interacts with the signal sequence on the polypeptide to induce translocation of the peptide across the plasma membrane and into the extracellular environment.

(Choices A and C) Mitochondria produce cellular energy in the form of ATP, and the SER functions to synthesize lipids, metabolize carbohydrates, and aid in detoxification. Neither of these membrane-bound organelles is present in prokaryotes.

(Choice D) Although *eukaryotic* protein synthesis of secreted proteins (eg, in ponyfish) may direct proteins from free ribosomes in the cytoplasm to the rough ER (RER) and then to the Golgi body prior to secretion, the question asks about SOD secretion in *P. leiognathi*. *P. leiognathi* is a prokaryote that *does not have* membrane-bound organelles like the RER and Golgi.

Educational objective:
Prokaryotic cells lack membrane-bound organelles and use specialized channels in the plasma membrane to secrete proteins. In contrast, eukaryotic cells contain membrane-bound organelles (eg, nucleus, Golgi body, mitochondria); the rough endoplasmic reticulum and Golgi body are involved in eukaryotic protein secretion.

Time spent:0 sec	**QID**:400784
Subject:	**System:**
Biology	Cellular Biology

Eukaryotic organelles such as mitochondria and chloroplasts are believed to be relics of formerly free-living prokaryotes. The transition from a hypoxic (low O_2) to an oxic atmosphere (21% O_2) is said to have enabled primitive eukaryotic anaerobes to engulf ancient aerobic prokaryotes and consequently acquire the ability to produce energy through oxidative phosphorylation. This endosymbiotic theory of eukaryotic evolution also postulates that endosymbiosis resulted in larger eukaryotic genomes, which originated from the partial transfer of mitochondrial genes to the nuclear genome. On integration into the host genome, mitochondria-derived genes became indistinguishable from the original nuclear genes.

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SODs are antioxidant enzymes that serve as the cell's first line of defense against reactive oxygen species (ROS). ROS produced by the electron transport chain damage proteins by oxidizing amino acid residues and metal ions on prosthetic groups but can accumulate during times of biochemical and environmental stress. Superoxide (O_2^-) radicals, a form of ROS, are sequestered by SODs and converted into less toxic hydrogen peroxide (H_2O_2) and O_2 gas.

Ponyfish cells containing *P. leiognathi* symbionts were exposed to a spindle fiber toxin that specifically inhibits microtubule polymerization. Given this information, which of the following would most likely result as a consequence of toxin exposure?

- A. *P. leiognathi* daughter cells with multiple copies of the Cu/Zn SOD gene [11%]
- B. Ponyfish daughter cells containing the same copy number of the Cu/Zn SOD gene [6%]

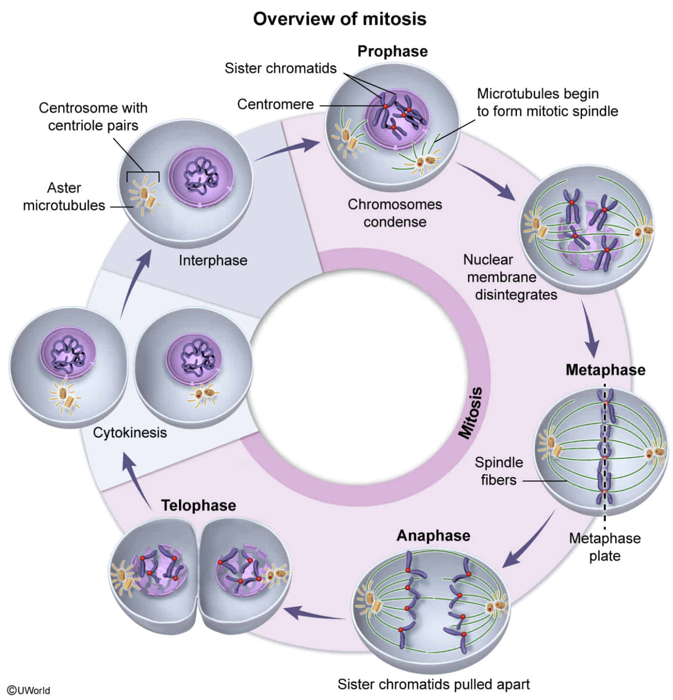
◯ C. Delayed separation of *P. leiognathi* cells during binary fission [19%]

✔ ◉ D. Nondisjunction in somatic ponyfish cells undergoing nuclear division [62%]

Correct

61% answered correctly

Explanation:



Somatic (non-sex) eukaryotic cells exist in two stages: interphase, during which the cell grows and synthesizes DNA; and mitosis, during which the nucleus divides to yield two identical daughter cells (with equal genetic material). The phases of mitosis include the following:

1. **Prophase**: Previously synthesized DNA condenses into chromatids. Two sister (ie, identical) chromatids are then connected by a kinetochore to form a single chromosome. The nuclear envelope disintegrates to allow mitosis in the cytosol. The mitotic spindle begins to form when centrosomes (microtubule-organizing centers consisting of a pair of centrioles) start migrating to opposite poles of the cell and **microtubule** filaments grow from them, resulting in spindle fibers.

2. **Metaphase**: The mitotic spindle forms fully when centrosomes align completely at opposite poles of the cell. Chromosomes align on a

central plane (ie, the metaphase plate) equidistant from each centrosome and attach to the spindle fiber microtubules via their kinetochores.

3. **Anaphase**: Spindle fibers pull sister chromatids apart (*disjunction*) while retracting toward the centrioles on opposite poles.

4. **Telophase**: The nuclear envelope reforms around the two sets of chromosomes, which return to their less compact, premitotic state. During this stage, the parental cell undergoes cytokinesis, the cytoplasmic division into two daughter cells.

Nondisjunction in mitosis or meiosis can occur when sister chromatids fail to separate properly during anaphase. Exposing eukaryotic **ponyfish** cells to

a spindle fiber toxin interferes with the formation of the spindle fibers, and therefore impairs the separation of sister chromatids. Because the sister chromatids cannot separate effectively, **nondisjunction during mitosis is expected**. As a consequence, affected ponyfish cells would *not* show an equal distribution of genetic material (genes) because nondisjunction can result in one daughter cell having a different number of chromosomes than the other **(Choice B)**.

(Choices A and C) The spindle fiber toxin would not influence the copy number of Cu/SOD genes or delay the separation of daughter cells in *P. leiognathi* (ie, a prokaryote) because prokaryotes divide by binary fission, a simple method of asexual reproduction that results in two identical daughter cells and does not involve spindle fibers. Therefore, a toxin that targets *eukaryotic* microtubules is unlikely to adversely affect *prokaryotic* cell division.

Educational objective:
Most eukaryotic cells (except germ cells) undergo cell division via mitosis. In contrast, prokaryotic cells duplicate via binary fission, a simple form of reproduction that does not involve the separation of chromosomes by spindle fibers.

Time spent: 0 sec	**QID:** 400785
Subject: Biology	**System:** Cellular Biology

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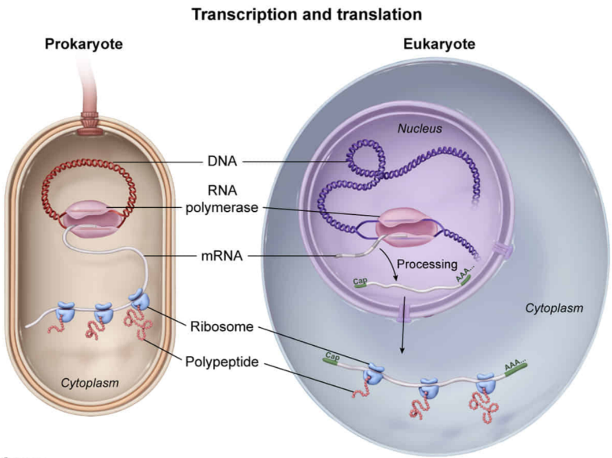
Investigators used a ribosome profiling technique to measure the level of SOD protein synthesis in *P. leiognathi* and ponyfish cells during ROS accumulation. After introduction of ROS, investigators found that *P. leiognathi* cells accumulated ribosome/SOD mRNA complexes at a faster rate than did ponyfish cells. What is the best explanation for this result?

- A. *P. leiognathi* SOD mRNA is capable of being bound by more than one ribosome. [9%]
- B. *P. leiognathi* SOD mRNA has a greater affinity for ribosomes than ponyfish mRNA. [22%]
- C. mRNA transcribed from the ponyfish SOD gene undergoes immediate degradation. [6%]
- ✔ D. mRNA transcribed from the bacterial SOD gene is being translated simultaneously. [62%]

Correct
61% answered correctly

Explanation:



Prokaryotic cells have no nucleus or nuclear envelope, and due to the lack of this physical barrier between DNA and the ribosomal machinery, **transcription and translation occur simultaneously** in the cytoplasm. Ribosome profiling was used to measure the level of SOD translation by detecting the number of SOD mRNA/ribosome complexes found within ponyfish (a eukaryote) and ***P. leiognathi*** **(a prokaryote)**.

The question states that *P. leiognathi* cells have a greater number of ribosome/SOD mRNA complexes compared to ponyfish cells. This rapid protein synthesis is likely due to the fact that all **SOD mRNA** transcripts in *P. leiognathi* can **quickly interact with ribosomes** even **before being fully transcribed** (ie, transcription and translation are coupled).

(Choice A) *Both* eukaryotic and prokaryotic mRNAs can be bound by multiple ribosomes in the cytoplasm.

(Choice B) Ponyfish mRNA is translated by the 80S ribosome whereas *P. leiognathi* mRNA is translated by the 70S ribosome. In this scenario, no information is given to suggest that ribosome affinity is a factor for SOD synthesis.

(Choice C) In eukaryotic (eg, ponyfish) cells, transcription of mRNA and its modification (eg, addition of the 5′ cap, 3′ poly-A tail, splicing) typically occur in the nucleus. The modifications, which result in a mature mRNA molecule, increase the stability of the transcript and prevent its degradation in the cytoplasm. Because the nucleus is separated from the cytoplasm by a nuclear envelope, mRNA must first be transported through nuclear pores to the cytoplasm, where it is translated by ribosomes.

Educational objective:
Prokaryotic cells have no nucleus; therefore, transcription and translation occur simultaneously in the cytoplasm (ie, translation begins before the mRNA is fully transcribed). In contrast, in eukaryotic cells transcription and post-transcriptional modifications occur in the nucleus but translation occurs in the cytoplasm.

Time spent: 0 sec	**QID**:400786
Subject: Biology	**System:** Cellular Biology

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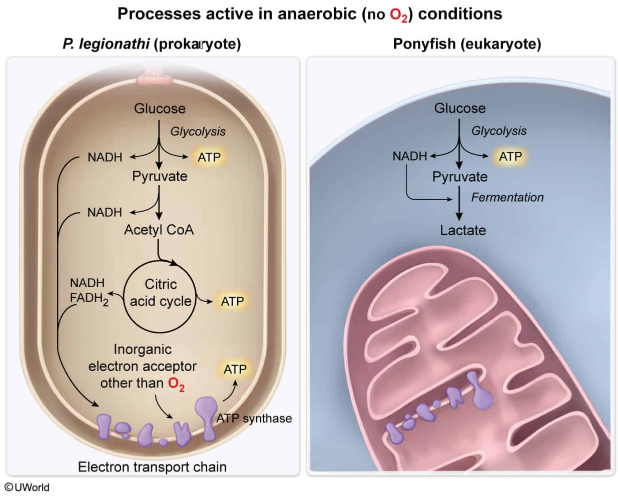
In anaerobic environments, *P. leiognathi* can produce energy by using an inorganic final electron acceptor other than oxygen in the electron transport chain. Under anaerobic conditions, which of the following is active in *P. leiognathi* but is NOT active in ponyfish cells?

- A. Glycolysis [4%]
- B. Gluconeogenesis [3%]
- C. Fermentation [32%]
- ✔ D. ATP synthase [59%]

Correct
58% answered correctly

Explanation:



Various organisms catabolize glucose, the primary fuel source for ATP production, via either cellular respiration or fermentation. Cellular respiration yields carbon dioxide and water whereas fermentation also yields waste compounds like lactate or ethanol. Both biochemical processes begin with glycolysis, a metabolic pathway that does not require O_2 and occurs under both aerobic and anaerobic conditions.

In glycolysis, a single molecule of glucose is broken down into two pyruvate molecules in the cytosol of both eukaryotes and prokaryotes (eg, glycolysis is *active in both P. leiognathi* and ponyfish cells under anaerobic conditions) **(Choice A)**.

In eukaryotes, cellular respiration continues when pyruvate is imported into the mitochondria and converted into acetyl-CoA, which then enters the Krebs cycle (citric acid cycle) to generate the necessary electron carriers (NADH and $FADH_2$) for the electron transport chain (ETC). ETC protein complexes pump protons across the inner mitochondrial membrane, generating a proton gradient with O_2 as the terminal electron acceptor. In comparison, prokaryotes have no mitochondria; instead, the ETC is located on the plasma membrane.

The enzyme ATP synthase harnesses the potential energy of the ETC-generated proton gradient to generate ATP from ADP and inorganic phosphate. It is stated in the question that ***P. leiognathi* can utilize the ETC in anaerobic environments** because it can use compounds other than O_2 as the final electron acceptor. Although ATP synthase does not directly need O_2 to generate ATP, the formation of the proton gradient by which it functions is **oxygen-dependent in ponyfish** cells but not in *P. leiognathi*. Consequently, under anaerobic conditions, ATP synthase is active in *P. leiognathi* but not active in ponyfish cells.

(Choices B and C) Fermentation metabolizes pyruvate under anaerobic conditions to oxidize NADH to NAD^+. Fermentation yields lactic acid in the muscles of ponyfish and alcohol in bacteria. Gluconeogenesis is the process by which noncarbohydrate carbon sources are converted into glucose. Because neither process requires O_2, fermentation and gluconeogenesis are active in both *P. leiognathi* and ponyfish cells.

Educational objective:

The Krebs cycle and the electron transport chain are only active in the presence of a final electron acceptor, such as oxygen (aerobic respiration) or inorganic ions (anaerobic respiration).

Time spent:0 sec	**QID**:400787
Subject: Biology	**System:** Cellular Biology